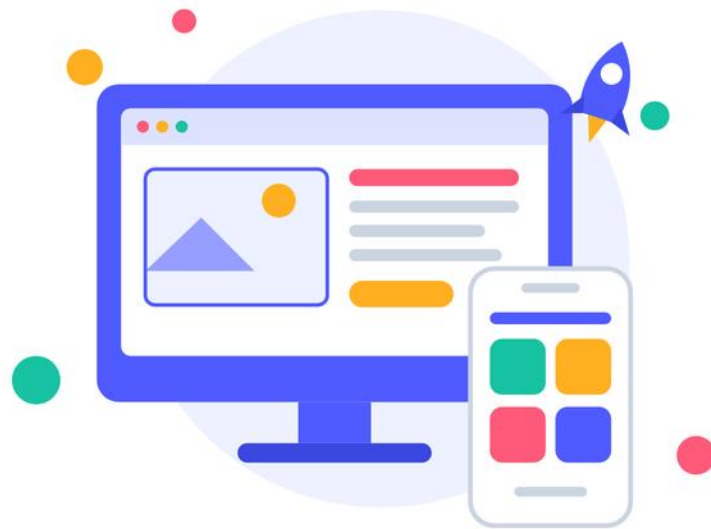


WEEK 02 · REFERENCE

THE CSS CATALOGUE

The companion to the lecture, and the one you keep open. Thirteen slides you will still be using in week 9, when a component needs a state and you want the selector in front of you rather than in your memory.

YOSSI ARVIV · FRONT-END WEB DEVELOPMENT · RUPPIN



EVERY SELECTOR, ON ONE PAGE

BASIC

`*` · `p` · `.class` · `#id`

ATTRIBUTE

`[type]` · `[type='email']` ·
`[href^='https']` ·
`[href$='.pdf']` · `[class*='card']`

LOGICAL

`:is()` · `:where()` · `:not()` · `:has()`

COMBINATORS

`A B` · `A > B` · `A + B` · `A ~ B`

STATE

`:hover` · `:focus` · `:focus-visible` ·
`:active` · `:disabled` · `:checked` ·
`:required` · `:invalid`

PSEUDO-ELEMENTS

`::before` · `::after` ·
`::first-line` · `::marker` ·
`::selection` · `::placeholder`

GROUPING

`h1`, `h2`, `h3`

POSITION

`:first-child` · `:last-child` ·
`:nth-child(2n)` · `:first-of-type` ·
`:only-child` · `:empty`

SPECIFICITY, WORKED

```
* (0,0,0)
p (0,0,1)
nav p (0,0,2)
nav > ul > li (0,0,3)
.lede (0,1,0)
p.lede (0,1,1)
[type='email'] (0,1,0)
input:required (0,1,1)
a:hover (0,1,1)
.nav .item a (0,2,1)
#page-title (1,0,0)
#nav .item a (1,1,1)
p::first-line (0,0,2)
:where(.card) p (0,0,1)
:is(.a, #b) p (1,0,1)
```

a

#id

b

.class · [attr] · :pseudo-class

c

element · ::pseudo-element

Compared column by column, like a version number. **b** never carries into **a**, however many of them there are.

:where() is always zero — the way to ship a default that anything can override.

THE CASCADE, IN ORDER

1 Origin

browser stylesheet, then yours. Yours wins.

2 Importance

`!important` reverses the origin order.

3 Inline

`style=""` beats any selector — this is the one place placement matters.

4 Specificity

the triple, column by column.

5 Source order

last one wins. Only reached on an exact tie.

Internal versus external is NOT a step. A `<style>` block and a linked file are decided by source order, like any other tie.

WHAT INHERITS, AND WHAT DOES NOT

INHERITS

color font font-family
font-size font-weight
line-height letter-spacing
text-align text-transform
white-space visibility
cursor list-style direction

DOES NOT

margin padding border
background width height
display position overflow
box-sizing opacity
transition

FORCE IT EITHER WAY

inherit take the parent's value.
initial the specification's default.
unset inherit if it inherits,
otherwise initial.
revert back to the browser's own
stylesheet.

UNITS

```
rem  x the ROOT font-size
em   x THIS element's font-size
%    of the containing block
ch   width of one "0"
ex   height of a lowercase x
vw   1% of viewport width
vh   1% of viewport height
vmin the smaller of vw and vh
px   one CSS pixel
fr   a share of free space (grid)
```

rem

type and spacing. Your default.

em

values that scale with this
element's own text.

%

widths.

ch

measure, and nothing else.

px

borders, outlines, small radii.

vw / vh

viewport-sized things. Sparingly.

COLOUR

```
color: #4c1eb8;           /* hex, 6 digits      */
color: #4c1eb880;         /* hex, 8 digits: + alpha */
color: #abc;              /* shorthand for #aabbcc */
color: rgb(76 30 184);    /* modern, space-separated */
color: rgb(76 30 184 / 50%); /* with alpha          */
color: hsl(258 72% 42%);  /* hue, saturation, light */
color: hsl(258 72% 42% / 0.5); /* with alpha          */
color: rebeccapurple;     /* 148 named colours exist */
color: currentColor;      /* whatever `color` is here */
color: transparent;
```

`currentColor` is the useful one nobody teaches: a border that always matches the text without repeating the value.

CONTRAST, AND WHERE TO CHECK IT

AA · 4.5:1

Body text. The threshold that matters, and the one most designs fail.

AA · 3:1

Large text — 24px, or 18.66px bold — and meaningful UI boundaries.

AAA · 7:1

Rarely required, always welcome.

webaim.org/resources/contrastchecker · color.adobe.com · colorhunt.co

DevTools shows the ratio in the colour picker itself.



TYPE

```
/* the system stack: installed, tuned, downloads nothing */
font-family: system-ui, -apple-system, 'Segoe UI', Roboto, Arial, sans-serif;

/* always end with a generic family */
font-family: Georgia, 'Times New Roman', serif;
font-family: ui-monospace, 'SF Mono', Menlo, Consolas, monospace;

/* the shorthand: style weight size/line-height family – size and family required */
font: 700 1.25rem/1.2 var(--font-sans);

/* self-hosted, because a CDN is a dependency you did not choose */
@font-face {
  font-family: 'Heebo';
  src: url('/fonts/heebo.woff2') format('woff2');
  font-display: swap;
}
```

THE BOX, AND ITS SHORTHANDS

```
margin: 16px;                /* all four */
margin: 16px 24px;           /* vertical | horizontal */
margin: 8px 16px 24px;       /* top | horizontal | bottom */
margin: 8px 16px 24px 32px;  /* top right bottom left, clockwise */
margin: 0 auto;              /* centre a block with a max-width */

padding: 16px;               /* same four patterns */

border: 1px solid hsl(220 20% 80%); /* width style colour */
border-bottom: 2px solid;           /* colour: currentColor */
border-radius: 12px;                /* four corners */
border-radius: 12px 12px 0 0;       /* TL TR BR BL */

outline: 3px solid;              /* like a border, but takes NO space */
outline-offset: 2px;

box-sizing: border-box;          /* width includes padding and border */
```

TOKENS: THE CONVENTION THIS COURSE USES

```
:root {  
  --color-bg: hsl(258 40% 99%);      /* the page ground      */  
  --color-surface: hsl(258 38% 96%); /* raised things      */  
  --color-border: hsl(258 20% 80%);  
  --color-text: hsl(258 30% 12%);  
  --color-text-muted: hsl(258 14% 36%);  
  --color-accent: hsl(258 72% 42%);  
  --color-accent-strong: hsl(258 76% 30%); /* the hover state */  
  
  --space-1: 0.25rem; --space-4: 1rem;  
  --space-2: 0.5rem;  --space-5: 1.5rem;  
  --space-3: 0.75rem; --space-6: 2rem;  
  
  --radius-sm: 6px;    --radius-md: 12px;    --radius-full: 999px;  
  
  --font-sans: system-ui, sans-serif;  
  --font-size-base: 1.0625rem;  --font-size-lg: 1.33rem;  
  --font-size-sm: 0.85rem;      --font-size-xl: 1.66rem;  
  
  --duration: 150ms;  
}
```

THE FIVE THAT WILL COST YOU TIME

WRONG

Specificity is a count of selectors.

`em` everywhere, including nested lists.

`width` is the width of the box.

`px` for type, so it never scales.

`!important` to win an argument.

RIGHT

Three numbers, compared column by column.

`rem` by default; `em` only when it should scale with this element.

Only with `box-sizing: border-box`. Otherwise padding and border are added on.

`rem`, so the reader's own setting is respected.

Lower the winner instead of raising the loser.

WHERE TO LOOK THINGS UP

MDN

`developer.mozilla.org` — the reference. Search `mdn css box-sizing` and take the first result.

CAN I USE

`caniuse.com` — is this safe to ship yet. Almost always yes, in 2027.

YOUR OWN DEVTOOLS

The fastest reference you own. Elements, then Computed, and every value the page actually resolved to — with the rule that set it.

Not Stack Overflow first. An answer from 2013 will tell you to use a float.